THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 9: **Light - Reflection and Refraction** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A magnifying glass of power 10 D is used. Its focal length and the maximum object distance (for a virtual image) are 1

(a) 1 cm; less than 1 cm **(b)** 10 cm; more than 10 cm
(c) 100 cm; less than 100 cm **(d)** 10 cm; less than 10 cm

2. Assertion (A): Concave mirrors are used in solar furnaces. Reason (R): Parallel rays of the Sun converge at the focus after reflection. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. A convex lens is used as a magnifying glass. The object must be placed 1

(a) between the optical centre and the focus **(b)** at 2F
(c) beyond 2F **(d)** at infinity

4. If a convex lens is cut along its principal axis into two halves, each half has 1

(a) half the focal length **(b)** the same focal length as the original
(c) double the focal length **(d)** zero

5. Assertion (A): A concave mirror always forms a magnified image. Reason (R): A concave mirror has a negative focal length. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A convex lens forms a real, inverted image smaller than the object when the object is placed 1

(a) beyond 2F **(b)** between F and 2F
(c) between F and the optical centre **(d)** at F

7. A diverging beam of light from a point source strikes a convex lens placed at distance f from the source. The refracted beam is 1

(a) parallel **(b)** convergent **(c)** scattered **(d)** divergent

8. Assertion (A): The speed of light changes when it goes from air to glass. Reason (R): The frequency of light also changes in glass. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. **Assertion (A):** The power of a concave lens is negative. 1
Reason (R): The focal length of a concave lens is taken as negative.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A convex lens of focal length 0.5 m has a power of +2 D. 1
Reason (R): The SI unit of power of a lens is the dioptre.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Two plane mirrors are at right angles. How many images are formed? Give reason. 2

12. A swimming pool of depth 3 m appears shallower. Find the apparent depth for n = 1.33 (nearly). 2

13. Light of frequency 5×10^{14} Hz travels from air into glass of n = 1.5. Find its speed and the frequency in glass. 2

14. A concave mirror cannot give a diminished virtual image. Explain. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A fish is at 1.5 m below the surface of a lake (n = 4/3). Find its apparent depth and explain. 3

16. How will the image formed by a convex lens change if its upper half is covered with a black paper? Explain. 3

17. An object is placed 15 cm in front of a convex lens and 15 cm in front of a concave lens, both of |f| = 15 cm. Compare the images. 3

OR

A concave mirror and a convex lens of the same focal length are used to form images of an object at 2f. Compare the nature, size and the side of the image. 3

18. Calculate the focal length of a lens that forms an erect image three times the size of an object kept 10 cm from it. 3

OR

How can you identify a concave mirror, a convex mirror and a plane mirror by looking at your face from a small distance? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A ray of light travelling in air strikes a glass slab. The refractive index of the glass is 1.5 and the speed of light in air is 3×10^8 m/s. The sine of the angle of incidence is found to be 0.75. 5
(i) Find the speed of light in glass.
(ii) Does the ray bend towards or away from the normal in glass?
(iii) State Snell's law.
(iv) Find the sine of the angle of refraction and the angle of refraction.

20. A student draws a table for a convex lens with u = 15, 20, 30, 40, 60 cm and f = 20 cm. Compute v for u = 30 and u = 40 and u = 60 and describe the trend. 5

OR

A concave mirror of radius of curvature 60 cm is used to obtain images of an object 6 cm tall placed at 45 cm and at 20 cm from the mirror. Find the image position, size and nature in each case. 5

END OF QUESTION PAPER — **Total: 40 Marks**